he’s going to see the same thing that Alice did. In the language of textbook quantum mechanics, Alice’s measurement collapses the wave function onto one of the two basis states, leaving Bob with a deterministic outcome. (In Many-Worlds language, Alice’s measurement branches the wave function, creating two different Bobs, each of whom will get a certain outcome.) That’s entanglement in action.

In the aftermath of the 1927 Solvay Conference, Einstein remained convinced that quantum mechanics, especially as interpreted by the Copenhagen school, did a very good job at making predictions for experimental outcomes, but fell well short as a complete theory of the physical world. His concerns were finally written up for publication in 1935 with his collaborators Boris Podolsky and Nathan Rosen, in a paper that is universally known as simply *EPR*. Einstein later said that the primary ideas had been his, Rosen had done the calculations, and Podolsky had done much of the writing.

EPR considered the position and momentum of two particles moving in opposite directions, but it’s easier for us to talk about qubits. Consider two spins that are in the entangled state written above. (It’s very easy to create such a state in the lab.) Alice stays home with her qubit, but Bob takes his and embarks on a long journey—say, he jumps in a rocket ship and flies to Alpha Centauri, four light-years away. The entanglement between two particles doesn’t fade away as they are moved apart; as long as neither Alice nor Bob measures the spins of their qubits, the overall quantum state will remain the same.

Once Bob arrives safely at Alpha Centauri, Alice finally does measure the spin of her particle, along an agreed-upon vertical axis. Before that measurement, we were completely unsure what such an observation would reveal for her spin, and likewise for Bob’s. Let’s suppose that Alice observes spin-up. Then, by the rules of quantum mechanics, we immediately know that Bob will also observe spin-up, whenever he gets around to doing a measurement.